Written in English for clarity — some technical terms don't translate naturally yet

One day of effort

Dataset for Article-Level Quoting (182 Million Pairs)

A question-answer resource generated from parsed regulations, engineered to facilitate reliable and citation-aware LLMs in Bahasa Indonesia.

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Bridging the Legal Comprehension Gap

- **The Problem**: LLMs often struggle with legal quoting and contextual comprehension because their training data lacks direct, grounded mappings from a question to the specific legal article that answers it.
- **The Solution**: I generated this massive Question-Answer (QA) dataset where every answer is an individual article or clause from the parsed legal hierarchy (ID_REG_Parsed).
- **Goal**: To facilitate LLMs that can not only answer legal questions but also cite the exact legal basis—a critical need in legal tech.
- **Disclaimer**: This dataset is intended for research and development only. QA pairs are synthetically generated from publicly available legal text and may not reflect official legal interpretations.

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Corpus vs. QA Pair Dataset

In legal NLP research, two major dataset types define the foundation of model capability — corpus datasets and QA-pair datasets.

Aspect	Corpus Dataset	QA Pair Dataset
Definition	A raw or **minimally processed** text collection (e.g., regulations, articles, clauses)	**Structured data linking** a question to its corresponding answer (e.g., article-level text)
Objective	Provide **general language understanding** and context exposure	Enable **specific** reasoning, retrieval, and answer generation
Structure	**Unlabeled**, paragraph- or document-level text	Explicit input–output mappings (**Question → Answer**)
Model Use Case	**Pretraining** and embedding learning.	**Fine-tuning**, evaluation, and supervised instruction learning
Example	*Pasal 3: Pembentukan peraturan perundang-undangan harus berdasarkan asas kejelasan tujuan...*	*Q: Apa asas pembentukan peraturan perundang-undangan menurut Pasal 3? → A: Asas kejelasan tujuan...*

At its core, communication with an LLM is a sequence of **questions from the user and answers from the model**. Training on a structured QA dataset ensures the model learns to **provide accurate, article-level responses**, aligning legal knowledge with real user queries.

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Strategic Choice for Quoting Accuracy

Metric	Template-Driven Synthesis	LLM Synthetic Generation
Generation Speed	**Extremely Fast**: Optimized script achieves near-instantaneous generation and transfer velocity	**Slow & Rate-Limited**: Subject to external API speed and high latency for millions of calls
Direct Cost	**Near-Zero Cost**: Leveraged free Kaggle Cloud CPU resources	**Prohibitively High Cost**: Requires paying per token for millions of generations (**could easily cost** thousands of dollars)
Quoting Accuracy	**Guaranteed Article Grounding**: The answer is structurally tied to the source article text	**Hallucination Risk**: High risk of generating incorrect or misquoted answers
Contextual Focus	Focuses on **direct quoting** and article comprehension	Often provides summarization or analysis, **not the desired direct**, citable article

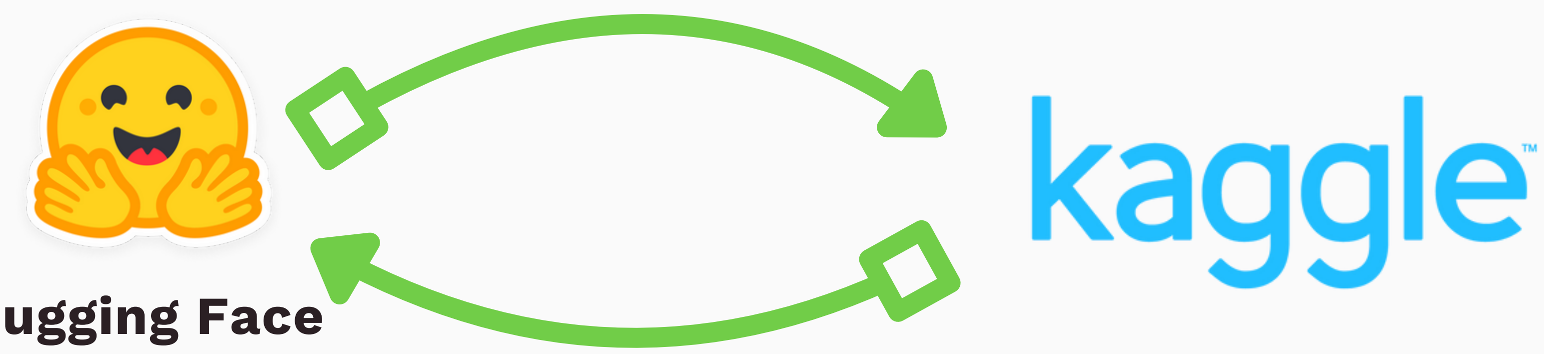
- **The Choice**: I chose Template-Driven Synthesis over generic Synthetic Generation (using a large external LLM like GPT) for maximum control and efficiency.
- **Process**: I used 50 predefined QA templates to automatically construct questions and answers based on the structured legal text from the ID_REG_Parsed repository.

Scaling QA Generation Without Scaling Cost

I conducted two controlled generation runs using the same base dataset but different template scales.

Metric	Small Version	Large Version
QA Templates	10	50
Rows Generated	33 Millions	182 Millions
Runtime	~20 minutes	~60 minutes

- Both runs were executed on a **single Kaggle instance** (which is free) using a cloud-to-cloud pipeline.
- The datasets were **pulled directly from Hugging Face, processed within the Kaggle notebook environment**, and pushed back to Hugging Face upon completion.
- **Small version repo**: Azzindani/ID_REG_QA_Small
- **Large version repo**: Azzindani/ID_REG_QA_Large



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Unlocking the Next Layer of Legal Understanding

- This project marks a major step toward transforming unstructured legal text into structured, machine-understandable data.

- From **250,000 Indonesian regulations**, we derived **182 million structured QA pairs** — a foundation for training models that truly understand and cite the law.

- The process shows that synthetic QA generation can be both scalable and cost-efficient, even within free cloud environments, demonstrating the feasibility of large-scale legal data curation for open research.

- **Key Application**:

  1. **Supervised Fine-Tuning (SFT)**: Utilize this QA dataset to train LLMs that respond to legal questions with clarity, structure, and source alignment.
  2. **Domain Adaptation** for Legal Context – Adapt general-purpose LLMs (e.g., Mistral, Gemma, or Llama) to Indonesian legal semantics.

- **Acknowledgements**:

  1. Hugging Face
  2. Kaggle

Curious about building high-quality datasets efficiently?

Let’s discuss ideas and approaches – Friendly and open, emphasizes sharing knowledge

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